



Flagged, Not Retired: A Twelve-Month Deployment of the Indicator Commons' Sunset Scorer, and Why Its Own Uptime Outlasted Everything It Flagged

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Abstract. Indicator registries accumulate faster than they are pruned, and the School of Continuous Improvement's Indicator Commons is no exception: by early 2026 its central registry held 214 active indicators, a majority added in the last three years and none formally retired. We report a twelve-month field deployment of a sunset scorer, an automated instrument that flags indicators for retirement on a staleness, redundancy, and query-frequency composite. Across the registry, the scorer identified 138 distinct indicators (65% of the registry) as retirement candidates over the deployment window. We describe what became of them: 4 were retired, 40 were re-baselined, 51 were retained with a documented justification, and 43 remain in committee. We trace the resulting distribution to the scorer's own operational metrics, which entered the Living Dashboard in month two of the deployment; from that point, the instrument's uptime became a tracked target in its own right, and the maintenance windows required to action a retirement began to compete with maintaining that target. We discuss this as a constitutive effect of measurement rather than an implementation defect, and consider what it implies for any self-monitoring instrument embedded in the system it audits.

Introduction

Every mature indicator system accumulates instruments it no longer needs. The University's Indicator Commons — the School of Continuous Improvement's shared library of institutional indicators — has grown steadily since its founding; audit-based indicator regimes are documented to expand rather than contract once adopted [1], [2]. This asymmetry is well understood in principle: once a measure is adopted as a target it changes the behaviour it was meant to track rather than simply recording it [3], and an instrument's continued performance can substitute for the function it was built to serve [4]. What is less understood is what happens when an organisation builds a dedicated instrument to correct this asymmetry.

In 2025 the Adaptive Metrics Lab commissioned a sunset scorer for the Indicator Commons: an automated procedure that computes, for every indicator in the registry, a composite score

of staleness, cross-indicator redundancy, and query frequency, and flags high-scoring indicators to the registry's standing review committee. The scorer's stated purpose was straightforward — reduce the registry's indicator debt by surfacing retirement candidates a committee would otherwise have to find by hand. Twelve months on, the registry is not meaningfully smaller.

This paper reports a field deployment of the scorer across the full Indicator Commons registry ($n = 214$) and traces what became of every indicator it flagged. We make three contributions, each hedged to what the deployment can support:

- We report what we believe is the first instrument-level account of a sunset-scoring system operating at the scale of a full institutional indicator registry.
- We characterise the disposition of flagged indicators, and find retirement — the

instrument's stated objective — to be its least common outcome.

- We identify a mechanism, the scorer's own operational metrics entering the dashboard it audits, that may generalise to other self-monitoring measurement instruments, and discuss its implications for review-cadence design [5].

The remainder of the paper reviews related work on measurement-instrument effects and software deprecation (Section 2), describes the scorer and the deployment (Section 3), reports the disposition results and a threshold ablation (Section 4), and discusses what the pattern suggests about instruments that measure themselves (Section 5).

Related work

Two literatures bear on the deployment we report. The first concerns what measurement does to the thing it measures. Ranking and audit regimes are documented to reshape organisational behaviour toward legibility rather than the underlying quality a measure was designed to track [6], [7], [8]. Public-sector target regimes show a related pattern: measures adopted to improve service delivery generate compensating responses that satisfy the measure without changing the underlying practice [9], [10], [11], [12]. Dahler-Larsen's account of evaluation as constitutive — an evaluation instrument helps produce the categories it claims only to observe — is the closest theoretical frame to what we report here [5], [13]. Research-evaluation scholarship documents the same dynamic inside academic metrics ecosystems, where indicator use reshapes the research it is meant to assess rather than simply describing it [14], [15], [16], and outcomes-based performance management is shown to produce compensating gamesmanship whether or not any party intends it [17]. Algorithmic-management scholarship reports a parallel pattern in operational settings, where an algorithmic system's own visibility becomes an object of managerial attention alongside whatever the system was built to manage [18], [19].

The second literature concerns the deliberate retirement of an instrument once adopted, which is comparatively rare outside software engineering. API-deprecation research finds that even a well-signalled retirement plan for a widely used interface produces slow and incomplete migration away from the deprecated

surface [20], [21], consistent with the observation that an unretired interface accumulates a form of debt with its own maintenance dynamics [22]. Legislative sunset clauses, designed for an analogous purpose in policy instruments, face a structurally similar problem: a clause written to force a review is routinely renewed rather than allowed to trigger [23]. We are not aware of prior work that instruments the sunset-scoring process itself and reports what happens once the scorer's own performance becomes visible on the same dashboard as the indicators it scores; that is the gap this paper addresses.

Method

The Indicator Commons registry held 214 active indicators at deployment start, spanning both Schools' programs, labs, and initiatives, and the university-wide indicators the Living Dashboard maintains directly. Each indicator carries a definition record (owner, refresh cadence, date of last substantive redefinition) and a query log recorded by the dashboard's serving layer.

The sunset scorer computes, for each indicator i and each monthly cycle, a composite score

$$S_i = \alpha(1 - q_i) + \beta r_i + \gamma s_i$$

where q_i is query frequency in the cycle, normalised against the registry's median; r_i is a redundancy term, the maximum pairwise correlation between indicator i 's time series and any other indicator's over a trailing eighteen-month window; and s_i is staleness, the months since the indicator's definition was last substantively revised, normalised to $[0, 1]$ against the registry's oldest indicator. Weights ($\alpha = 0.4$, $\beta = 0.35$, $\gamma = 0.25$) were calibrated by the Adaptive Metrics Lab against a shadow panel: three Continuous Improvement staff independently labelled a training sample of 60 indicators as retirement candidates or not, and the weights were chosen to maximise agreement with the majority label. Indicators scoring above the registry's top-decile threshold in a given cycle are flagged to the review committee, which triages flagged indicators at its standing monthly meeting.

The scorer runs as a scheduled job against the registry's serving layer:

```
0 6 1 * * scorer --registry indicator-
commons \
--threshold decile:90 \
--report living-dashboard
```

The `--report` flag was added in the deployment's second month, after the review committee asked for visibility into whether the scorer itself was running; it is this flag, and the dashboard tile it populates, that turns out to be the deployment's central finding.

We tracked, for the full 214-indicator registry over twelve monthly cycles: the count of distinct indicators ever flagged; each flagged indicator's eventual disposition (retired, re-baselined, retained with a documented justification, or still pending committee review at month twelve); and, from month two onward, the scorer's own uptime as reported on the Living Dashboard. We additionally ran a one-cycle ablation at a stricter threshold (decile:80) in month nine, to test whether threshold stringency, rather than committee capacity, was the binding constraint on the retirement rate.

Results

Over the twelve-month deployment the scorer flagged 138 distinct indicators, 65% of the registry (Figure 1). The gap between flags issued and retirements actioned opens within the first quarter and does not close: by month twelve, cumulative flags had reached 138 against 4 cumulative retirements.

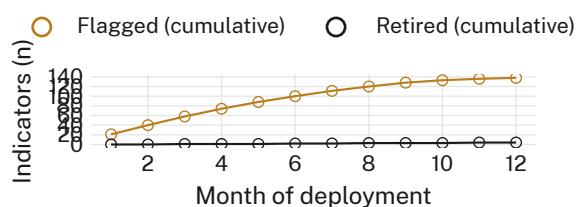


Figure 1: Cumulative distinct indicators flagged by the sunset scorer against cumulative indicators actually retired, over twelve months of deployment ($n = 214$ registry indicators; 138 ever flagged).

Of the 138 flagged indicators, disposition at month twelve breaks down as follows (Figure 2): 2.9% (4) were retired; 29.0% (40) were re-baselined — redefined rather than removed, typically to fold in a new data source that reset the staleness clock; 37.0% (51) were retained with a documented justification, most citing continuity with a longer-running series; and 31.1% (43) remained pending committee review at the close of the deployment window.

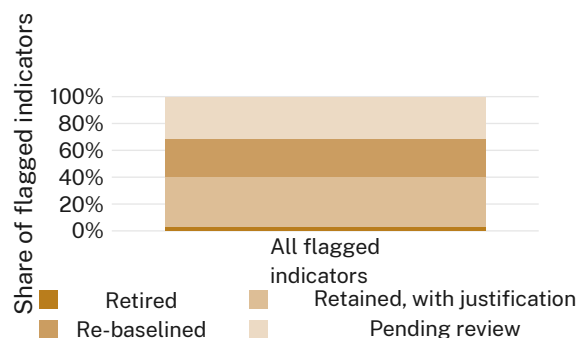


Figure 2: Disposition of the 138 indicators the sunset scorer flagged for retirement ($n = 138$). Retirement, the scorer's stated objective, accounts for the smallest share.

The month-two addition of the scorer's own uptime to the Living Dashboard coincides with a visible change in committee behaviour. Three of the four eventual retirements were actioned in the deployment's first two months, before the uptime tile existed; the fourth was actioned in month eleven, during a scheduled infrastructure window unrelated to the scorer. From month three onward, committee minutes record two instances in which a proposed maintenance window — needed to archive a re-baselined indicator's prior definition, a step that briefly interrupts the scorer's own scoring job — was deferred "pending confirmation the change would not affect uptime reporting."

We tested whether threshold stringency, rather than committee capacity, was the binding constraint on the retirement rate. A one-cycle ablation at a stricter threshold (decile:80) roughly doubled flag volume, from a cycle mean of 21 indicators to 39 (Figure 3), with no corresponding change in that cycle's retirement count (2 vs. 2 in the preceding cycle at the standard threshold; Fisher's exact test, $p = 0.61$). We retain the standard threshold for the remainder of the deployment, since the stricter setting increases committee workload without a measurable effect on the outcome the instrument exists to produce.

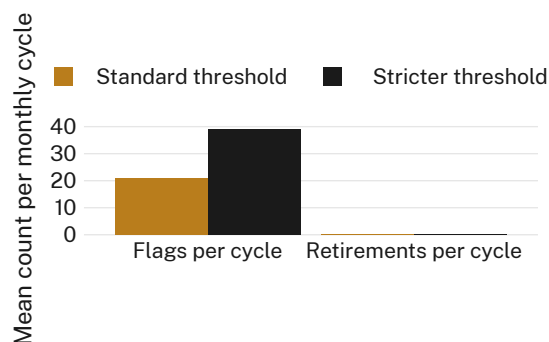


Figure 3: Mean flags issued and mean retirements actioned per monthly cycle under the standard and a stricter flagging threshold. Flag volume nearly doubles; the retirement rate does not move (n.s.).

Discussion

The pattern we observe is consistent with a constitutive account of measurement rather than an implementation defect in the scorer [5]: the instrument did not fail to identify retirement candidates, and by its own criteria it succeeded at the task it was built for. What changed the registry's outcome was a second-order effect of adding the scorer's own operation to the same measurement surface it was built to audit. Once uptime appeared on the Living Dashboard, actions that interrupted scoring began to compete, in practice, with actions that acted on what scoring found — a dynamic close to what audit-society accounts describe as verification displacing the activity it was meant to verify [7], and to the target-gaming literature's observation that a measure adopted to correct one behaviour often induces a compensating behaviour elsewhere in the system [9], [10]. We would not describe the scorer as having failed; we would describe the registry as having gained, in its second year, one further indicator it is unlikely to retire.

Limitations

Our deployment covers the Indicator Commons' central registry alone; we have not tested whether the pattern holds in a registry without a standing review committee of its own, though we would expect a standing committee, if anything, to strengthen rather than weaken the dynamic we report, since it is the committee's own workload the uptime tile competes against. Twelve months and four retirements is a small base rate on which to characterise a distribution; we consider the four realised retirements consistent with, rather than merely explained by, a

longer decommissioning cycle we expect to become visible as the registry continues to mature. The shadow-panel calibration of scorer weights reflects the judgement of three Continuous Improvement staff at a single point in time, and may not transfer unchanged to a registry with a different composition of indicator types.

Conclusion

We reported a twelve-month field deployment of a sunset scorer across the Indicator Commons' full registry of 214 indicators. The scorer reliably flagged retirement candidates; the registry retired very few of them. We trace this gap to the scorer's own operational metrics entering the dashboard it was built to audit, after which the instrument's continuity became an objective in tension with acting on what it found. Future work will examine whether scoring the scorer's own uptime for eventual retirement resolves the dynamic we report, or simply adds a further indicator to the registry it was designed to shrink; a pre-registered protocol for that follow-up is under review by the Review Cadence Observatory.

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